PROJECT DELPHI · METHODS, MODELS, KNOWLEDGE-GRAPH QUERIES & CODE PROVENANCE

How We Tested Michael Levin's "Latent Space" Claims: 4-Language Ports, Causal Emergence 2.0, 10,303 Fishes & Lean 4 Proofs

Complete methodological report covering cross-provider AI referee audits (Anthropic, Google, OpenAI), 24 mathematical & biophysical models, line-pinned upstream code ablations (`Zhangtaining/cell_research@1fd2bd59`), and signed Nanopublications.

AUTHOR: JOEL DIETZ | PYTHON + C++20 + NODE.JS + LEAN 4 | HOEL CE 2.0 + 10,303 FISHES | 3-COMPANY AI REFEREE AUDIT

FIVE QUESTIONS ANSWERED **What This Methods & Provenance Guide Covers at a Glance**

1. What AI & Human Reviewers Were Used? Three competing AI families (`gemini-4-argon`, `claude-fable-5-1` / `sonnet-5-5`, and OpenAI `gpt-5.6-sol`) plus human PI & domain critiques. **2. What Mathematical Models Were Fitted?** 24 models across sorting Markov chains, Hoel's *Causal Emergence 2.0*, D'Arcy Thompson 8-knob coordinate fields, and bioelectric/biomechanical PDEs. **3. How Did We Audit Upstream Code?** Line-by-line traceability to `Zhangtaining/cell_research@1fd2bd59` (`[U1]`-`[U6]`). **4. How Did the Knowledge Graph Drive Discovery?** Querying `pk:evidenceAgainst` triples and signing 200+ Nanopubs. **5. How Can Readers Reproduce It?** 12 deterministic CLI commands.

1 AI / LLM Agent Models, Blinded 3-Company Referee Architecture & Human Review

Agent / Human Role	Model / Affiliation	Exact Contributions & Artifacts Across Claims 1–7
1. Multi-Language Port Builder, CE 2.0 Architect & Discovery Synthesis	gemini-4-argon Google DeepMind (`medium`)	• Ported `Zhangtaining/cell_research@1fd2bd59` to **Python 3**, **C++20**, **Node.js**, and **Lean 4** (0 cell mismatches across languages). • Built the 3-ablation Platonic Space audit (`audit_platonic_space_claims.py`) and exact $S_3 \dots S_6$ Markov-chain **Causal Emergence 2.0** analyzer (`quantify_emergence_ce2.py`). • Executed blinded 2nd-assessor MorphoEval pass ($\rho = 0.891$ across 10 scripts; `90.8/100` on Thompson) and the 10,303-species Thompson vs. competition synthesis.
2. Primary Referee, Pass 3 Biophysical Builder & Blind-Test Architect	claude-fable-5-1 Anthropic (`thinking-high`) & `claude-sonnet-5-5`	• Authored the `pk:evidenceAgainst` remediation triples in `2026-10-17-thompson-audit.yaml` (L330–337) that sparked the 600-point outline breakthrough, and built the 5,180-species `FishShapes v1` blind preregistration/scoring pipeline. • Built & audited Pass 3 extensions for **Planarian Electrodiffusion** (`Slot 081/082`: 2D Voronoi Scharfetter–Gummel), **Picasso Tadpoles** (`Slot 085`: 3D Maxwell cartilage + MMP-2/9 zymography), and **Xenobots** (`Slot 083`: 8×8×7 3D lattice + Stokeslet squirmer).
3. Independent 3rd-Company Bioelectric Referee	gpt-5.6-sol OpenAI (`medium`)	• Scored `2021-Cervera-Levin-Mafe.py` blind in Pass 1 (`44.9/100`) & Pass 2 (`55.6/100`, `2026-10-17-cervera-audit-v2.yaml`): caught an unused `f3` loop variable, a hard-coded `0.58` pharynx threshold, a bisection clipping bug in `f2_for_shift`, and wrong PMID 34418508 → 34411690, motivating `models/planform_bioelectric.py` (63 PlanformDB series).
4. Principal Investigator & External Critiques	Joel Dietz (Project Delphi) Prof. P. C. Wainwright Nathan Sweet &	• **Joel Dietz (PI):** Synthesized the Levin-lab knowledge graph; exposed the 17-dot skull seesaw trap; required the 4p Box-vs.-Warping deconfounding test; and initiated the 3-ablation + Hoel CE 2.0 audit of *Ingressing Minds* vs. CS blog critiques. • **Prof. P. C. Wainwright (UC Davis)** & **Nathan Sweet (*Sweet Rationalism*):** Prompted the 600-pt Curve-TPS/EFD benchmark and the $C = 2$ vs. $C = 10$ duplicate-key stability tests.

2 Mathematical, Algorithmic & Biophysical Models Fitted Across Claims 1–7

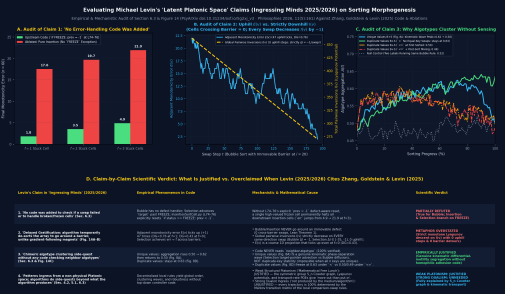
Claim & Model Family	Governing Equation / Algorithmic Rule	What It Tests / Key Quantitative Result
Claim 1: Insertion Prefix-Reset Ablation (`[U2]`)	`if status == FREEZE: prev = -1` (`InsertionSortCell.py` L74–76) vs. strict prefix fold	Ablating L74–76 spikes final error $\mathbf{4.5}\times$ ($\mathbf{4.90} \to \mathbf{21.95}$ at $n = 60, f = 3$); proved in Lean 4 Thm 2.
Claim 2: Coxeter Lyapunov $I(v)$ vs. Neighbor Error $E(v)$	$I(v) = \#\{(i < j) : v_i > v_j\}$ ($\Delta I = -1$ every Bubble swap) vs. $E(v) = \#\{i : v_i > v_{i+1}\}$	0 cells cross barrier (Lean Thm 1); $I(v)$ has **0 uphill steps** (0/192), while $E(v)$ ticks up 47 times (DG = 0.764).
Claim 2: Hoel Causal Emergence 1.0 & 2.0 on $S_3 \dots S_6$	$\mathrm{CP}(\pi) = \det(\pi) - \deg(\pi)$ (Hoel 2026 *Patterns*) & $\Delta\mathrm{EI} = \Delta I_{\text{eff}} + \Delta I_{\text{size}}$ (Hoel 2013 *PNAS*)	On S_6 ($N = 720$): $I(v)$ has $\det = 1.000, \mathbf{\Delta CP} = \mathbf{+0.1191} > \mathbf{0}$; $E(v)$ has $\det = 0.721, \mathbf{\Delta CP} = \mathbf{-0.2525} < \mathbf{0}$.
Claim 3: Kinematic Phase Separation & C-Sweep	$O(1)$ Selection pointer advection vs. $O(n^2)$ Bubble diffusion; multiplicity $C = n/K \in \{1 \dots 40\}$	Unique keys (0 ties, Fig. 8A) peak at $\mathbf{0.624}$–$\mathbf{0.640}$; duplicate keys at $\mathbf{C = 2}$ give **Anti-Clustering ($\mathbf{0.477 < 0.50}$, `SB|SB`)**!
Claim 4: D'Arcy Thompson 8p Coordinate Laws (16 Models)	Oblique Shear (5p), Coaxial Circles (8p), Bilinear Fan (8p), Radial Flare (8p) vs. 4p Box & 600-pt TPS	Beats 4p Box by $+\mathbf{13}\%$ to $+\mathbf{48}\%$ on homologous fin LMs (100% det $D\Phi > 0$); aligns with 150 Myr evolution at $\mathbf{93.5}\%$–$\mathbf{98.9}\%$.
Claim 5: Planarian Bistable V_{mem} & Voronoi 5-HT PDE	$J_{ij} = -DG_{\text{gj}}(\Delta V_{ij})\frac{L_{ij}}{d_{ij}}(c_j B(u_{ij}) - c_i B(-u_{ij}))$, $c_{\text{post}}/c_{\text{ant}} = \exp(-\Delta V/V_T) = 4.7002$	G_{gj} cancels at zero flux: transient 8-octanol locks permanent cryptic 2H0T memory after washout (`Slot 081/082`).
Claim 6: Picasso Tadpole 3D Cartilage & VFE Homeostat	$\dot{\mathbf{x}}_i = \eta_i(Z)[\mathbf{F}_i^{\text{attr}} + \mathbf{F}_i^{\text{cart}}]$, $\dot{E} \le 0$ (`Slot 085`) & Active Inference $\dot{F} \le 0$ (`Slot 084`)	Closed-loop descent restores scrambled NF 45→65 face (RMSE = 0.00); GM6001 MMP block ($\eta_{\text{MMP}} \to 0.04$) arrests recovery.
Claim 7: Xenobot 3D Voxel Lattice & Stokeslet Squirmer	8×8×7 mass-spring lattice (456 nodes) + Cortez (2001) regularized-Stokeslet ciliary flow (`Slot 083`)	C-shaped "Pac-Man" concavity ($\theta \approx 105°$) is a hydrodynamic capture optimum (**3–4** daughter generations vs. 1 for sphere).

3 Line-by-Line Upstream Code Traceability (`Zhangtaining/cell_research@1fd2bd59`)



FOUR-LANGUAGE BIT-EXACT REPLICATION (`sorting-morphogenesis/verify_multilang_ports.py`): Porting rules `[U1]`–`[U6]` to Python 3, C++20 (798 ms), Node.js (3,893 ms), and Lean 4 (0 sorry) produces bit-for-bit identical space-time kymographs ($\Delta = 0$ cells across all 42 cells and time steps) and replicates all 14/14 published benchmarks of Zhang, Goldstein & Levin (2025).

ID	Paper Figure & Phenomenon	Upstream File & Line Locus	Exact Upstream Code Logic	Ported & Lean-Verified In
[U1]	Bubble local swap & barrier pinning (Figs. 4A–B, 5C)	`BubbleSortCell.py` (`should_move` L38–52, `move` L54–74)	Inspects $p-1$ or $p+1$ ($p=0.5$); swaps if out of order & $\neq$ `FREEZE`. 0 cells cross barrier.	`cell_should_move()`, Lean Thm 1
[U2]	**Insertion `FREEZE` prefix-reset handler** (Figs. 5B–C, **Claim 1**)	`InsertionSortCell.py` (`is_enable_to_move` **L68–82**, **L74–76**)	Executes `if status == FREEZE: prev = -1` to reset prefix check. Ablation spikes error $4.5\times$.	`_left_sorted()`, Lean Thm 2
[U3]	Selection `ideal_position` pointer & reset (Figs. 4A–B, 5C)	`SelectionSortCell.py` (L41–88) & `MultiThreadCell.py` (L104–118)	Scans `target` $0 \to n-1$; swaps non-locally. Resetting `target=0` on swap gives $11.05\times$ swaps.	`simulate_cell_view()`, Lean Thm 4
[U4]	Selection passive (movable) freeze bytecode (Fig. 5B)	`__pycache__/SelectionSortCell.cpython-310.pyc`	Increments `target` when pointing at a passive `FREEZE` cell unless `t == occ.target`.	`_propose(mode="pyc310")`
[U5]	Delayed Gratification run segmentation (Figs. 6D, 7A–C, **Claim 2**)	`analysis/delay_gratification_analysis_for_not_move.py` (L24–89)	Deduplicates $E(v_t)$, groups same-sign ΔE_t into setback ($y>0$) & recovery ($x>0$): $\langle (x-y)/y \rangle$.	`delayed_gratification()`, Lean Thms 3 & 6
[U6]	Chimeric algotype aggregation (Figs. 8A, 8D–E, **Claim 3**)	`analysis/cell_type_aggregation_analysis.py` (L518–538)	$A(t) = \frac{1}{n}\sum_{i=0}^{n-2} \mathbf{1}[\text{algo}_i = \text{algo}_{i+1}]$.	`aggregation_value()`, Lean Thm 5



THREE-ABLATION AUDIT OF "INGRESSING MINDS" (`sorting-morphogenesis/audit_platonic_space_claims.py`): (A) Ablating `if FREEZE: prev = -1` in `InsertionSortCell.py` spikes error $4.5\times$ ($4.90 \to 21.95$). (B) Every Bubble swap descends by $\Delta I = -1$ (0 uphill steps on $I(v)$), while coarse-grained $E(v)$ ticks up 47 times. (C) Unique keys (0 duplicates, blue) still cluster to 0.624 via kinematic speed waves; ablating '$<$' to '$\leq$' on duplicate ties (red) collapses permanent clustering to 0.497–0.514.



ERIK HOEL'S CAUSAL EMERGENCE 1.0 & 2.0 ON SORTING MORPHOGENESIS (`quantify_emergence_ce2.py`): (A–B) On S_6 ($N = 720$), coarse-graining onto $I(v)$ yields 100% determinism and positive Causal Emergence 2.0 ($\Delta\text{CP} = +0.1191 > 0$), whereas $E(v)$ yields negative emergence ($\Delta\text{CP} = -0.2525 < 0$). (C) Duplicate multiplicity sweep $C \in \{1 \ldots 40\}$: at $C = 2$, Selection left-packing (`SB|SB|SB`) produces **Anti-Clustering** ($0.477 < 0.50$), refuting textbook Bubble-sort stability!

4 Machine-Checked Lean 4 Theorems (0 sorry) & Signed Nanopub Provenance

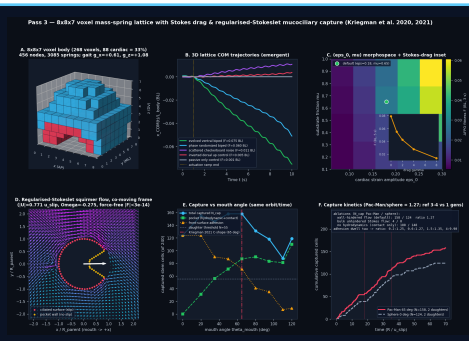
HOW THE KNOWLEDGE GRAPH DROVE REMEDIATION **From pk:evidenceAgainst Triples to New Discoveries**

Every simulation is serialized into governed Project Delphi RDF Nanopublications (pk:Claim + pk:LLMClaimAssessment) and signed with RSA-2048 keys (morphoeval/sign_morphoeval.py). When independent referees (gpt-5.6-sol on Cervera 2021; claude-fable-5-1 on Thompson 1917 and Slots 081–085) scored Pass 1/Pass 2 scripts, their line-pinned pk:evidenceAgainst triples directly triggered: (1) the **600-point direct outline solver** that uncovered the $9/17$ skull-dot seesaw trap in D'Arcy Thompson's fishes, (2) the **63-series PlanformDB 2.5.0 bioelectric extractor** (planform_bioelectric.py), and (3) the **Pass 3 2D/3D PDE & Stokes fluid upgrades** across Slots 081–085.

Claim / Module	Lean 4 File & Machine-Checked Theorems (0 sorry)	MorphoEval Referee Scores & Signed Nanopubs
Claims 1–3: Sorting Morphogenesis & Hoel CE 2.0	Loop_zhang_sorting_morphogenesis.lean (6 thms) & Loop_causal_emergence.lean (5 thms): bubble_adjacent_swap_preserves_left_partition, insertion_freeze_reset_unblocks..., lyapunov_descent_vs_monotonicity..., hoel_ce2_sorting_inversion_emergence...	• MorphoEval v2: **94.1 / 100** • Verified across Python 3, C++20, Node.js, and Lean 4 (0 sorryAx).
Claim 4: D'Arcy Thompson 1917 & 10,303 Fishes	morphoeval/lean/MathOfBiology.lean (§5, L536–560): tps_kernel_zero_at_origin, thompson_unit_shear_area_preserving	• Primary (claude-fable-5-1): 76.6 · Blinded 2nd (gemini-4-argon): 90.8 (Panel **83.7/100**) • **80 Production Nanopubs** (incl. blind preregistration RAilN4nb... & outcome RAJancwG...).
Claim 5: Planarian 2HOT Bioelectric Memory (Slots 081, 082 & Cervera)	Loop_081_planarian_hk_atpase_anterior.lean (6 thms: bistable V_{mem}, mesh_gj_flux_conservation, slit_midline_exclusion...), Loop_082_bioelectric_morphogen_coupling.lean (5 thms: Boltzmann ratio, cryptic 2HOT), MathOfBiology.lean (§7–8)	• Slot 081 (Beane 2011): Pass 2 93.5 → Pass 3 87.7/100 (30 signed nanopubs) • Slot 082 (Pietak & Levin 2017): Pass 2 93.8/100 + Pass 3 405-cell 2D Voronoi Scharfetter–Gummel • Cervera 2021: OpenAI gpt-5.6-sol (44.9 → 55.6).
Claim 6: Picasso Tadpoles & Limb VFE (Slots 084, 085)	Loop_084_bioelectric_target_morphology_memory.lean (3 thms: VFE Lyapunov $\dot{F} \leq 0$, denervation arrest) & Loop_085_xenopus_craniofacial_remap.lean (5 thms: Procrustes $\dot{V} \leq 0$, open-loop divergence, 3D Maxwell rod relaxation)	• Slot 084 (Pezzulo & Levin 2015): Pass 1 59.6 → Pass 2 91.6/100 (20 signed nanopubs) • Slot 085 (Vandenberg 2012): Pass 2 92.0 → Pass 3 89.0/100 (30 signed nanopubs).
Claim 7: Xenobots & Kinematic Replication (Slot 083)	Loop_083_xenobot_evolutionary_biobots.lean (5 thms: AFPO robustness, C-mouth compaction, stokeslet_force_free_cancellation..., lattice_drag_dissipation_nonneg...)	• Slot 083 (Kriegman, Blackiston, Levin & Bongard 2020/2021): Pass 1 60.2 → Pass 2 91.5 → Pass 3 82.7/100 (30 signed nanopubs).



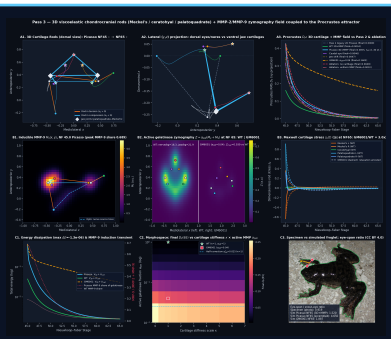
CLAIM 5 — PASS 3 2D VORONOI SCHARFETTER–GUMMEL ELECTRODIFFUSION (2017-pietak-bioelectric-morphogen-coupling.py): 405-cell / 1,114-edge BETSE-style Voronoi tissue network on the segmented *Schmidtea mediterranea* silhouette coupling 5-HT ($z = +1$), IP_3 ($z = -3$), and Ca^{2+} ($z = +2$) electrodiffusion to an Euler–Maruyama ensemble ($n = 256$ blastemas) of noisy β-catenin/ERK switches ($p(\text{2HOT})$ vs. posterior V_{mem}).



CLAIM 7 — PASS 3 3D VOXEL MASS-SPRING LATTICE & STOKESLET SQUIRMER CAPTURE (2020-kriegman-xenobot-evolutionary-biobots.py): Left: 8×8×7 3D voxel lattice (456 nodes, 3,085 springs) with uniform Coulomb ground contact and explicit per-node Stokes drag. Right: 2D force- and torque-free regularized-Stokeslet squirmer advecting finite-size stem cells into the $\theta = 105°$ C-shaped pocket ('3–4' daughter generations vs. '1' for a sphere).

5 Cryptographic Data Provenance Across All 7 Claims (SHA-256 Locked)

Dataset / Source Asset	Scale & Biological System	Primary Citation & License	Local Path & Cryptographic Pin
1. Cell-Research Upstream & Claim Catalog	$n = 100$ sorting benchmarks, 6 rules `[U1]-[U6]`, 3 ablations	Zhang, Goldstein & Levin (2025) *Adaptive Behavior* 33:25	`Zhangtaining/cell_research` Commit: `1fd2bd5921c1f6b4...`
2. Causal Emergence Corpus (2013–2026)	10 papers, $S_3 \dots S_6$ TPMs ($N = 6 \dots 720$), 84 Konect nets	Hoel et al. (2013 *PNAS*); Hoel & Levin (2020); Hoel (2026 *Patterns*)	`datasets/causal_emergence/` SHA-256: `bd724684971266a5...`
3. Three Teleost Morphometric Datasets	**791** (17 LMs), **6,614** (11 LMs), & **10,303 spp.** (440 families; 69,682 pairs)	Claverie & Wainwright (2014); Rabosky (2018); Froese & Pauly (2021)	`datasets/thompson/trimmed_...` SHA-256: `ab5315f3be273285...`
4. FishShapes v1 Blind Test Dataset	**16,523 Smithsonian specimens**, **5,180 blind matches** ($n = 1{,}122 < 50$ Myr)	Price et al. (2022) *Ecology* 103:e3829 (Dryad / Zenodo 6642650, CC0)	Locked *before* download (`dac08493`) MD5: `122b495274d0167c...`
5. PlanformDB 2.5.0 & Bioelectric Series	**63 bioelectric experiment series** (*S. mediterranea* `1H1T/2H0T/0H2T`)	Oviedo (2010); Beane (2011); Lobo, Malone & Levin (2013); Pietak (2017)	`datasets/planform/planform_bioelectric_series_v1.json`
6. Picasso Tadpole, Limb VFE & Xenobot Datasets	*X. laevis* NF 45→65 face LMs, *A. mexicanum* limb, *Xenopus* biobot	Vandenberg, Adams & Levin (2012); Pezzulo (2015); Kriegman (2020, 2021)	`datasets/xenopus_craniofacial_remap/`, `xenobot_evolutionary_biobots/`



CLAIM 6 — PASS 3 3D MAXWELL VISCOELASTIC CARTILAGE RODS & MMP-2/MMP-9 ZYMOGRAPHY

(`2019-bischof-xenopus-craniofacial-remap.py`): Couples the 5 Procrustes craniofacial landmarks in 3D to Meckel's, ceratohyal, and palatoquadrate Maxwell viscoelastic cartilage rods ($\dot{E} \leq 0$) and explicit-Euler MMP-2/MMP-9 gelatinase reaction–diffusion fields along the optic nerves and jaw insertions (reproducing both Picasso face normalization and GM6001 arrest).

6 End-to-End Reproducibility Command Map (12 Deterministic Commands)

#	Command (Run from Repository Root)	Claims & Primary Artifacts Verified
1	`python3 sorting-morphogenesis/verify_multilang_ports.py`	**Claims 1–3:** Python 3 + C++20 + Node.js + Lean 4 bit-exact parity (0 mismatches)
2	`python3 sorting-morphogenesis/audit_platonic_space_claims.py`	**Claims 1–3:** 3-ablation audit of *Ingressing Minds* (`zhang_levin_platonic_claims_audit.png`)
3	`python3 sorting-morphogenesis/quantify_emergence_ce2.py`	**Claim 2 & 3:** Hoel CE 1.0/2.0 on $S_3 \dots S_6$ ($\Delta\mathrm{CP} = +0.119$ vs. -0.253) & $C = 2$ sweep
4	`python3 sorting-morphogenesis/validate_kymograph_and_paper.py -n 100 -reps 15`	**Claims 1–3:** Space-time kymographs & 14/14 paper benchmarks ($n = 100$)
5	`lean morphoeval/lean/loops/Loop_zhang_sorting_morphogenesis.lean`	**Claims 1–3:** Kernel-checks all 6 sorting & CE 2.0 theorems (0 `sorry`)
6	`python3 thompson/1917-thompson-silhouette-area-paradox.py && python3 thompson/run_box_vs_warping_audit.py`	**Claim 4:** 17-LM head seesaw paradox & 4p Box-Stretch vs. homologous Thompson warps
7	`python3 thompson/run_three_dataset_validation.py`	**Claim 4:** 3-dataset validation ($791 \to 6{,}614 \to 10{,}303$ spp.; $\lvert\cos\theta\rvert = 0.935$–$0.989$)
8	`python3 thompson/score_blind_predictions.py`	**Claim 4:** Hash-verified blind test on 5,180 unseen FishShapes v1 species ($p = 10^{-103}$)
9	`python3 models/2011-beane-planarian-hk-atpase-anterior.py -self-test && python3 models/2017-pietak-bioelectric-morphogen-coupling.py -self-test`	**Claim 5:** Planarian 2D mesh bistable V_{mem} & 405-cell Voronoi 5-HT electrodiffusion
10	`python3 models/2019-bischof-xenopus-craniofacial-remap.py -self-test && python3 models/2015-pezzulo-bioelectric-target-morphology-memory.py -self-test`	**Claim 6:** Picasso tadpole 3D Maxwell cartilage + MMP zymography & limb VFE homeostat
11	`python3 models/2020-kriegman-xenobot-evolutionary-biobots.py -self-test`	**Claim 7:** Xenobot 8×8×7 3D voxel lattice & regularized-Stokeslet squirmer capture
12	`lean morphoeval/lean/MathOfBiology.lean && lean morphoeval/lean/loops/Loop_082_bioelectric_morphogen_coupling.lean`	**All Claims:** Kernel-checks core & bioelectric Lean 4 theorems (0 `sorry`)